



KOGOD SCHOOL *of* BUSINESS
AMERICAN UNIVERSITY • WASHINGTON, DC

Project Requirements Specifications

Project Name: Dell Consulting Project

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Communication Log

Date	Communication Topic	Comm Type	Participants		
			Team Members	Client Reps	Other
Deliverable 1					
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12/9	Presentation Preparation	Zoom	Jordan, Isaac, Ronald, & Caleb	N/A	N/A
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Executive Summary

Project Title:

Dell Consulting Project: Enhancing the Perpetual-to-Subscription (P2S) Software Licensing Process

The global Software market is on a remarkable growth trajectory, with projected revenue reaching US\$702.00bn by 2024 (Statista, 2024). Within this market, Enterprise Software is anticipated to dominate with a market volume of US\$294.30bn (Statista, 2024). The market is expected to grow at an annual rate of 5.01% from 2024 to 2029, reaching US\$896.20bn by 2029 (Statista, 2024). The United States is poised to lead globally, generating US\$363.40bn in revenue in 2024 (Statista, 2024). Amid this market growth, Dell Technologies is transitioning from perpetual to subscription-based licensing for third-party software to capitalize on this expansion. However, the shift introduces complexities in quoting, ordering, and managing royalties. Current manual processes are inefficient, costly, and prone to errors, affecting customer satisfaction and scalability. Our project addresses these challenges by designing an automated system to streamline operations and position Dell to better compete in this lucrative market. The existing quote-to-cash process relies on manual steps, creating bottlenecks in pricing, validation, and vendor integration. These inefficiencies hinder Dell's ability to scale and respond quickly to market demands. Our analysis proposes a comprehensive solution that automates these tasks, consolidates workflows, and integrates real-time data from third-party vendors. This will reduce lead times, improve accuracy, and elevate customer satisfaction, aligning with the market's anticipated growth. The recommended system incorporates an automated quoting tool, API-driven vendor integration, and contract management software. It supports key functionalities such as dynamic pricing, automated validation, and seamless order tracking. Non-functional requirements emphasize scalability, security, and usability, ensuring the system meets Dell's global operational needs. The system will manage essential data entities like customer profiles, vendor details, and product configurations. APIs will enable real-time data integration, ensuring accurate quotes and streamlined order processing. The data collected will also provide valuable insights for predictive analytics and strategic decision-making. This project aligns Dell's operations with market trends, leveraging automation to enhance efficiency and scalability while positioning the company to capture opportunities in a rapidly growing global Software market.

1. Introduction

In this document, we describe the requirements for the system designed to improve Dell Technologies' Perpetual-to-Subscription (P2S) Third-Party Software Product Licensing Partnerships. We begin by outlining the business case and system vision. Next, we provide a detailed discussion of the stakeholders involved in the process and the current business process related to software licensing. Following that, we describe the functional, non-functional, and data requirements of the system. For a glossary of terms and acronyms used in this document, please refer to Appendix A.

Project Concept

2. Business Case and System Vision

2.1 Client

Company Name: Dell Technologies

Department/Unit Involved: Infrastructure Transformation

Business Focus:

Dell Technologies is a major player in the technology industry, specializing in providing enterprise software and hardware solutions. The company is transitioning its business model from selling perpetual software licenses to a subscription-based model. This shift aligns with industry trends towards cloud-based services, where customers pay recurring fees to access software.

Client Activities:

- Software Licensing: Dell partners with third-party software vendors to integrate or resell their software as part of Dell's hardware offerings.
- Enterprise Solutions: Dell provides comprehensive solutions that include servers, laptops, accessories, and software, tailored to meet customer needs.
- Process Improvement: The company is focused on enhancing its quote-to-cash process for third-party software add-ons, aiming to reduce costs and improve efficiency.

Client Contact Role:

- Role/Title: Infrastructure Transformation Project Manager
- Responsibilities: Overseeing the transition to subscription licensing, managing partnerships with third-party vendors, and improving business processes related to software sales and fulfillment.

2.2 Business Case

With the shift from perpetual to subscription-based licensing, Dell faces new complexities in its deal-sizing, quoting, ordering, fulfillment, and royalty management processes for third-party software components. These processes currently depend heavily on manual steps, resulting in high operational costs, inefficiencies, and potential for error, which negatively impacts profitability and customer satisfaction.

The current multi-layered sales and fulfillment cycle requires seamless coordination to deliver tailored solutions. Starting with requirement gathering, Dell's sales team formulates a solution using Dell products and services and creates a "Bill of Materials" that includes third-party components when necessary. However, this manual effort in validation, pricing, and ordering has resulted in increased turnaround times and reduced scalability, especially as Dell's partner ecosystem grows and more third-party components are integrated.

The challenges within the existing process include:

- **Manual, time-consuming steps:** Reliance on manual processes for third-party integrations drives up operational costs and increases the risk of human error.
- **Extended lead times:** Generating and validating quotes is slowed by the need to manually aggregate information from various sources, resulting in delays that impact sales and customer experience.
- **Inflexibility in payment and partner terms:** Limited automation makes it challenging to adapt to dynamic pricing, diverse payment options, and evolving vendor relationships.

To address these challenges, we propose a comprehensive solution that incorporates automation, streamlined processes, and standardized vendor agreements. This will bring immediate and long-term benefits:

1. **Time Reduction and Accurate Quotes through Automation:** By automating deal configuration and quote generation with a quote automation tool, Dell can significantly reduce costs associated with manual processes, streamline approvals, and improve quote accuracy. This tool offers pre-set templates, pricing models, and approval workflows, enabling the sales team to generate detailed quotes faster and focus on building client relationships. Additionally, Dell will provide an API to third-party vendors, allowing them to perform quoting directly, further enhancing response times and ensuring compliance with partner terms. This automation tool improves efficiency, reduces lead times, and enhances the overall customer experience.

2. **Standardized Vendor Agreements through Contract Management Software:** Using contract management software, such as PandaDoc, Dell can create standardized agreements that reduce the need for customizations across vendors. Standardization will simplify compliance, support scalability, and maintain consistency across transactions.

Implementing these changes will lead to:

- **Increased efficiency and scalability:** Reducing manual processes can lower operational costs by up to 30% and reduce quote lead times by 20-30%.
- **Enhanced customer satisfaction:** Faster quote generation and reliable delivery times will strengthen Dell's reputation and customer relationships.
- **Stronger vendor relationships:** Standardized agreements streamline partner interactions and ensure mutually beneficial terms.

This initiative will not only optimize Dell's quote-to-cash process but also align with industry best practices, positioning Dell to meet the demands of a rapidly evolving software licensing landscape.

2.3 System vision

The recommended system is designed to streamline Dell Technologies' Perpetual-to-Subscription (P2S) third-party software licensing partnerships by automating core components of the quote-to-cash process. This solution incorporates an automated quoting system, an API-driven vendor integration, and contract management software. These elements will collectively reduce lead times, increase quote accuracy, and enhance Dell's ability to scale with third-party vendors efficiently. By consolidating quote generation, approval, and order alignment into a unified workflow, the system will enable faster, more reliable responses to customer needs while improving operational costs by reducing manual intervention.

3. Stakeholders

Dell's quote-to-cash process involves nine key stakeholders, each with a specific role impacted by the new system implementation:

- **Sales Account Representatives:** As the first point of customer contact, Sales Account Representatives will benefit from faster, more accurate quote generation tools. Automation allows them to focus more on understanding client needs and less on manual quoting tasks, which enhances customer satisfaction and drives sales efficiency.
- **Financial Analysts & Auditors:** Financial analysts will have better access to structured, real-time data on quote profitability, enabling more accurate assessment of bundled offerings with third-party software. Auditors will benefit from improved transparency and traceability in the system, simplifying financial reviews and compliance checks.
- **IT & Technical Support Specialists:** With an automated quote-to-cash system, IT & Technical Support will have a more streamlined infrastructure to manage, reducing technical overhead. The system's API functionality also makes it easier for IT to support third-party integrations, ensuring smooth implementation for Dell's products and partner software.
- **Order Processing Specialists:** Automated quote and order alignment will reduce discrepancies between customer quotes and fulfilled orders. Order Processing Specialists can confirm orders faster and with greater accuracy, leading to fewer order errors and enhanced operational efficiency.
- **Supply Chain Coordinators:** The streamlined quote-to-cash system will ensure accurate and timely updates on product requirements, allowing Supply Chain Coordinators to coordinate hardware assembly and software integration more effectively, which improves order accuracy and on-time delivery.
- **Partner Account Managers:** The new system will help Partner Account Managers by integrating third-party vendors via API, enabling partners to access real-time quoting information. This transparency supports smoother pricing adjustments and more flexible vendor relationships, ultimately improving vendor satisfaction and alignment.
- **Billing Specialists:** The automation in the quote-to-cash process will lead to faster, more accurate invoice generation based on fulfilled orders. Billing Specialists will experience fewer errors in invoicing, leading to improved payment cycles and clearer communication of payment terms.
- **Legal & Compliance Officers:** With standardized contract management tools, Legal & Compliance Officers will find it easier to manage and enforce compliance with industry regulations, particularly in third-party software licensing. This system minimizes risks associated with inconsistent terms and improves compliance tracking.

- **Customer Success Managers:** The enhanced system will streamline processes for tracking order fulfillment and customer satisfaction metrics, giving Customer Success Managers a clearer view of post-sale interactions. Faster resolution of technical or product-related issues will improve customer satisfaction, as they can proactively engage IT support when needed.

4. Business Process Requirements

4.1 Baseline Process

Dell's current quote-to-cash process for delivering customized solutions to customers involves multiple, detailed steps designed to ensure accuracy, compliance, and customer satisfaction. This process is complex, as it coordinates various internal and external stakeholders to deliver hardware, software, and third-party components seamlessly.

- **Requirement Gathering and Solution Design:** The process begins with Dell's Sales Account Representatives working closely with customers to identify specific needs. During this stage, the sales team gathers comprehensive requirements, such as technical specifications, desired features, budget constraints, and timeline expectations. Using this information, Sales Account Representatives create a tailored solution that may include Dell products and third-party software or hardware components.
- **Quote Generation and Approval:** Based on the customer's requirements, Dell generates a formal quote detailing the Bill of Materials (BOM), itemized pricing, applicable discounts, and payment terms. This quote, typically valid for 30 days, undergoes an internal approval process to ensure pricing accuracy and compliance with Dell's contractual agreements. For solutions involving third-party components, the sales team must confirm resale pricing from partners to finalize the BOM, ensuring that the quote aligns with all vendor agreements.
- **Customer Review and Order Confirmation:** Once the quote is issued, the customer reviews the BOM and terms. If the quote meets the customer's expectations, they submit a Purchase Order (PO) to confirm their commitment. Dell's order processing team then verifies the PO details against the approved quote to prevent any discrepancies before moving forward with order fulfillment.
- **Fulfillment Process:** Upon order confirmation, Dell's fulfillment process begins. This multi-stage process includes:
 - **Hardware Production:** Dell's supply chain team assembles the required hardware components.
 - **Software Installation:** Software, both Dell and third-party, is configured according to the customer's specifications.
 - **Shipping and Delivery:** The customized hardware and software bundle is packaged and shipped to the customer's specified location.

- **Licensing and Activation:** For software components, particularly third-party add-ons, Dell provides customers with software keys via a Licensing Portal. This step ensures that customers can activate and fully utilize all software elements in their custom solution.
- **Invoicing and Payment Collection:** After the solution is delivered, Dell generates and sends an invoice outlining the payment terms agreed upon in the quote. Billing Specialists track payments and apply them to the corresponding invoices, ensuring that records are kept accurate for both Dell's and partners' financial tracking.
- **Partner Coordination for Third-Party Components:** If the solution includes third-party products or services, Dell's Partner Account Managers work with partners to fulfill any financial obligations. This involves:
 - Determining whether a contractual payment plan is in place for each partner.
 - Processing payment to partners based on either royalties or negotiated terms.
 - Ensuring partner compliance and maintaining an accurate record of transactions for future audits.
- **Post-Delivery Support and Customer Success:** Once the product is delivered and invoiced, Customer Success Managers monitor customer satisfaction and address any issues related to the solution. In cases where customers experience technical issues, IT & Technical Support Specialists assist in ensuring product functionality and fulfillment of service agreements.

For further details on the baseline BPM, please refer to Appendix B1

4.2 Scope of Work

In this Project, we will specifically be looking at the process of obtaining the resale quote under the Quotation Team. This can take a significantly long time, hence impacting the customer's experience and sometimes leading the customer to opt for directly purchasing from the vendor.

The Quotation Team's functional roles include preparing the Bill of Materials (BOM), coordinating with the customer, and ensuring that the BOM is exactly what they expected.

This Team also coordinates with the Vendor to get the resale quotes before a final quote is issued to the customer.

4.3 Business Process Analysis

a) Business Process Improvement Goal(s)

Our primary goal is to improve operational efficiency by reducing manual processes in deal configuration and quote generation. This focus on automation aims to reduce time, streamline approvals, and ensure accurate quotes, addressing Dell's need for a faster, more reliable process. This goal directly supports Dell's objective of enhancing customer experience and

improving response times by enabling both Dell's sales team and third-party vendors to generate quotes quickly and accurately, in compliance with partnership terms.

b) Business Process Improvement Analysis

The current quotation process faces several significant challenges that hinder efficiency and effectiveness. Firstly, relying on manual steps for integrating third-party vendors and partners is a major bottleneck. This manual approach increases operational costs and heightens the risk of human error, which can lead to inaccuracies in quotes. Secondly, the process of generating and validating quotes is slowed by the need to manually gather and verify information from multiple sources. This results in extended lead times, negatively impacting sales cycles and customer satisfaction. Lastly, the process lacks flexibility in adapting to dynamic pricing and diverse payment terms due to limited automation, making it difficult to respond swiftly to market changes and customer needs.

Eliminating redundant manual tasks by automating data collection and integration with third-party systems would significantly reduce lead times and errors. Adding a step for automated validation of information could streamline the quote generation process. Merging steps like **P3.1**, **P3.2** with **P3.3** into a single automated pricing engine could simplify operations and enhance accuracy. Adding time stamps at the different points in the process from receiving requirements, creating quotations, and notifying the customer will support analytics in measuring the efficiency of the process, and providing timely insights into pricing trends and customer preferences.

4.4 Target Process

Automation and Integration:

Automate Data Collection: Implement an automated system to gather and integrate data from third-party vendors and internal sources. This will reduce manual effort, minimize errors, and speed up the process.

Automate Pricing and Validation: Use an automated pricing engine that consolidates "Determine price and quantity" and "Use Dell Price" into a single step. This engine should dynamically adjust prices based on real-time data from vendors.

Streamlined Workflow:

Merge Steps: Combine steps P3.1 (Determine price and quantity) P3.2 (Use Dell Price) and P3.3(obtaining resale quotes) into a single automated task to streamline operations. After the salesperson has determined the customer requirements, they will enter them into the system, which based on the quantities entered will determine the prices. The system

will have an option of choosing the 3rd party vendor, and based on details submitted, it will automatically set the prices and generate a quote. All the third-party vendors have to be linked to this billing and quotation system through APIs.

Reorder Steps: Adjust the sequence to ensure that all necessary data is collected before creating quotes, reducing back-and-forth communication.

Enhanced Flexibility:

Dynamic Payment Options: Integrate flexible payment processing that can adapt to various customer needs, enhancing customer satisfaction.

Improved Analytics:

Data Collection for Analytics: Introduce steps to collect additional data points such as customer preferences and transaction timestamps. This data can be used for predictive analytics to enhance decision-making.

These improvements aim to reduce lead times, lower operational costs, and enhance customer satisfaction by providing faster and more accurate quotes. Enhanced analytics capabilities will offer insights into customer behavior and market trends, supporting strategic decision-making.

5. System Actors

Actors are all entities that interact with the system. Actors can be human (i.e., users) or external systems that need to interact with the system. They are categorized as either Primary Actors, which have goals that the system needs to fulfill, or Secondary Actors, which provide necessary support but do not have primary goals in the system. Below is a list of each actor's purpose and functionality in the system. Complete actor cards with more details can be found in Appendix C.

Following this process, this section requires the following:

5.1. Primary Actors

1. Customer Goals:

- Receiving and Confirming the Bill of Materials
- Submitting the purchase order
- Make the payment

2. Sales Executive

Goals:

- Creating a Bill of Materials
- Sending The Bill of Materials to the customer

3. Quotation Executive

Goals:

- Creating a Quote
- Sending customer Quote
- Sending order to the customer

4. Billing Executive

Goals:

- Placing Orders
- Creating customer invoices
- Sending invoices to the customer
- Tracking third-party components
- Notifying customers of payments that are due.

5. Fulfillment Executive

Goal:

- Receive order placements

5.2. Secondary Actors

6. Quotation Management System

Goal: N/A

6. Functional Requirements

The system is designed to streamline Dell's quote-to-cash process by consolidating each step into an efficient, automated workflow. From the initial BOM generation to order

fulfillment, the system reduces manual processes by automating price validation, quote generation, and order tracking. It integrates with third-party vendor systems through APIs, ensuring that accurate, real-time data is available for generating quotes and invoices. This enables the Sales, Quotation, and Billing Executives to serve customers faster and with fewer errors. The use case diagram in Appendix D visually represents these functional interactions, while the specific use cases, detailed in Appendix E, describe the workflow steps and system actions needed to support each actor's goals.

7. Non-Functional Requirements

7.1 Look and Feel Requirements

- The user interface should be intuitive and align with Dell's brand guidelines, ensuring a consistent experience across internal and external systems.
- The interface should provide quick access to key functions such as quote generation and order tracking.

7.2 Usability Requirements

- Users should be able to perform primary tasks (quote generation, order placement) within 3 clicks or fewer.
- Training materials should be accessible to new users, and they should be able to perform key tasks within one day of training.

7.3 Performance Requirements

- The system should process and generate quotes within 5 seconds, reducing lead times significantly.
- It must support a minimum of 100,000 transactions per day to accommodate scalability.

7.4 Operational Requirements

- The system must be available 24/7, with scheduled maintenance windows only outside of business hours.
- Data backup should occur every 4 hours, with a disaster recovery time of under 30 minutes.

7.5 Maintainability and Portability Requirements

- The system should support remote access for Dell employees, allowing secure connections from any location.
- Regular updates to support third-party software integration must be possible without significant downtime.

7.6 Security Requirements

- User authentication should include two-factor authentication for internal users.
- Data encryption must be applied to all customer and vendor data, ensuring compliance with Dell's data protection standards.

8. Mandated Constraints

- The system must integrate with Dell's existing infrastructure and utilize their preferred Oracle DBMS.
- All interactions with third-party vendor systems must comply with the legal agreements in place with each vendor, particularly concerning data access and sharing.
- The solution must be deployed on Dell's private cloud infrastructure, ensuring adherence to internal data handling policies.

9. Relevant Facts and Assumptions

9.1. Facts

- Dell Technologies has a global presence and requires multi-currency and multi-language support.
- The system will operate within Dell's established technology infrastructure and should integrate with existing customer management and billing systems.

9.2. Assumptions

- All third-party vendors involved in the quote-to-cash process will provide API access to relevant data.
- The system's users will have basic technical skills, reducing the need for extensive support materials.

- Dell will provide ongoing feedback on system functionality to ensure continuous improvement.

Appendix A

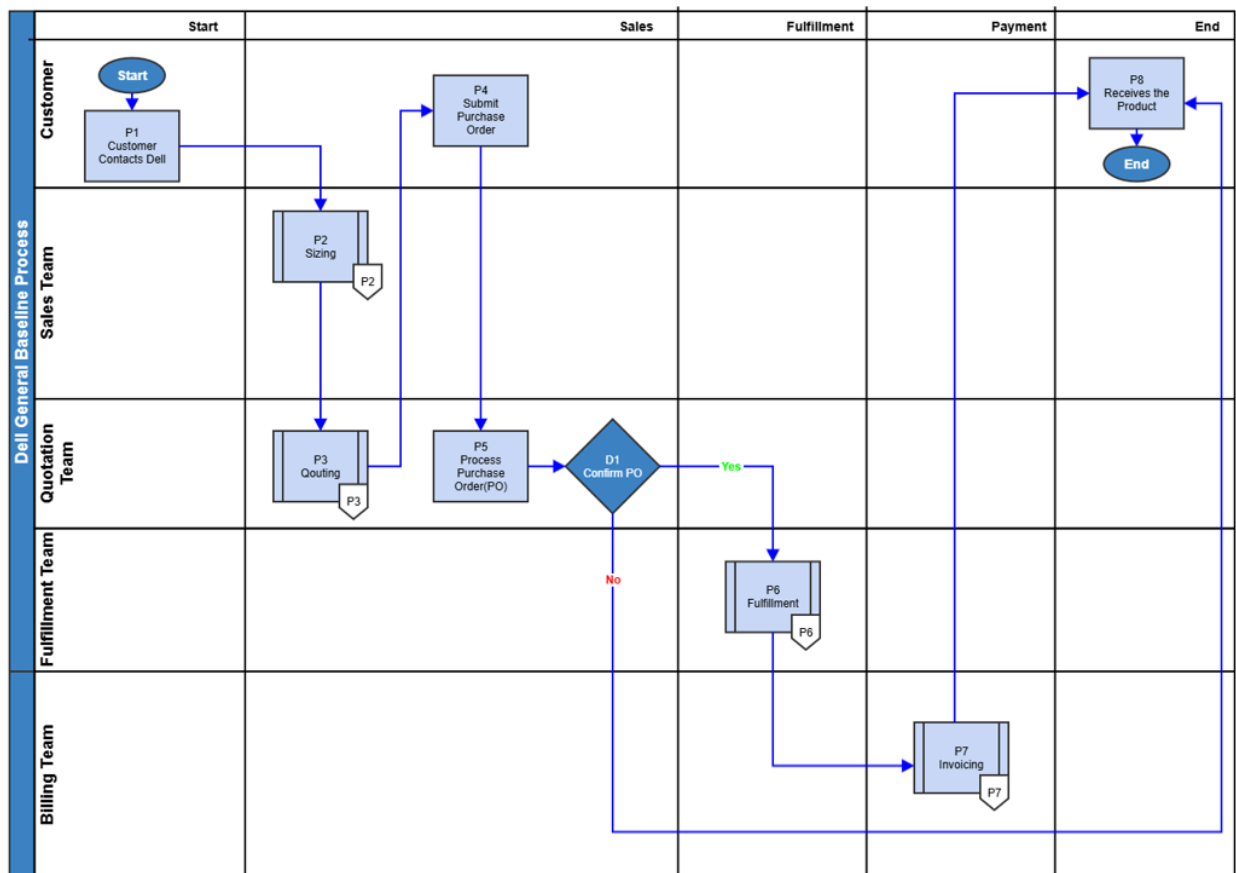
Project Glossary

- Bill of Materials (BOM): A detailed list of hardware, software, and components needed to fulfill a customer order. It includes specifications, quantities, and associated vendors.
- Quote-to-Cash (QTC): The complete end-to-end business process that covers quoting, order placement, invoicing, and payment processing. The QTC process is critical for revenue generation and customer relationship management.
- Sales Executive: A primary actor in the system responsible for initiating the sales process by generating a Bill of Materials and ensuring customer requirements are met.
- Quotation Executive: A system user responsible for generating and sending quotes to customers based on BOM specifications.
- Billing Executive: The individual who oversees processing customer orders, generating invoices, and confirming payments within the system.
- Fulfillment Executive: The role in charge of ensuring customer orders are fulfilled by coordinating the shipment or installation of purchased products and services.
- Purchase Order (PO): A document submitted by the customer as a formal acceptance of a quote, authorizing Dell to proceed with order fulfillment.
- Third-Party Vendor: External companies providing hardware or software components that Dell resells or integrates into their customer solutions.
- Quotation Management System (QMS): A system used to automate and streamline the quoting process, integrating with external vendors and validating pricing and availability.
- Invoice: A document generated by the Billing Executive and sent to the customer, detailing the costs associated with the products and services provided, based on the purchase order.
- System Actor: Any user or external system that interacts with the quote-to-cash process automation system, either directly (primary actors) or in a support role (secondary actors).
- Business Process Modeling (BPM): A method of representing business processes through diagrams to identify and analyze workflows and interactions among different actors.
- Use Case: A description of a specific function or feature of the system, including the steps and interactions required to achieve a particular business goal.
- Perpetual-to-Subscription (P2S): A business model transition where traditionally perpetual licenses for software are converted to subscription-based licensing, offering continuous updates and support.
- System Integration: The process of linking the automated system with third-party systems to enable data exchange, such as updating pricing or tracking inventory in real-time.
- API (Application Programming Interface): A set of protocols that allow different systems (e.g., the automated quote system and third-party vendor systems) to communicate and share data securely.
- User Authentication: Security processes, such as two-factor authentication, are required for system users to verify their identity before accessing system features.

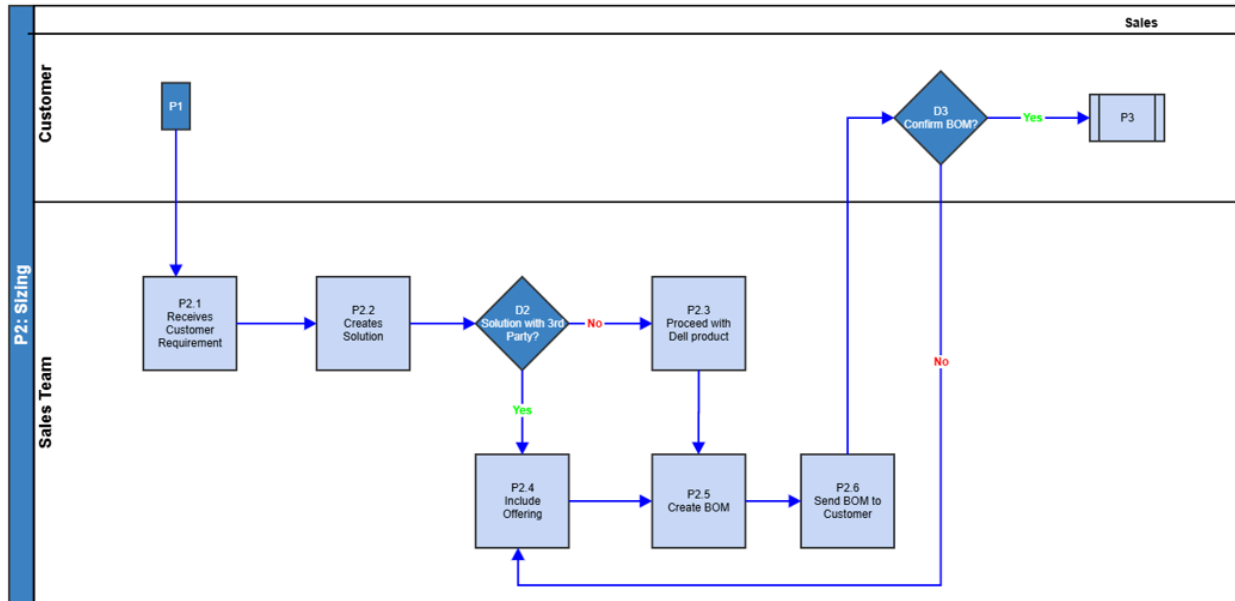
- **Data Encryption:** A security measure that protects sensitive information by converting it into a coded format, ensuring data privacy and compliance with legal standards.
- **Non-Functional Requirements:** Specifications that define the quality attributes of the system, including performance, usability, security, and maintainability.
- **Mandated Constraints:** Specific restrictions or requirements, such as technology choices or regulatory standards, that the system must adhere to during development.

Appendix B1

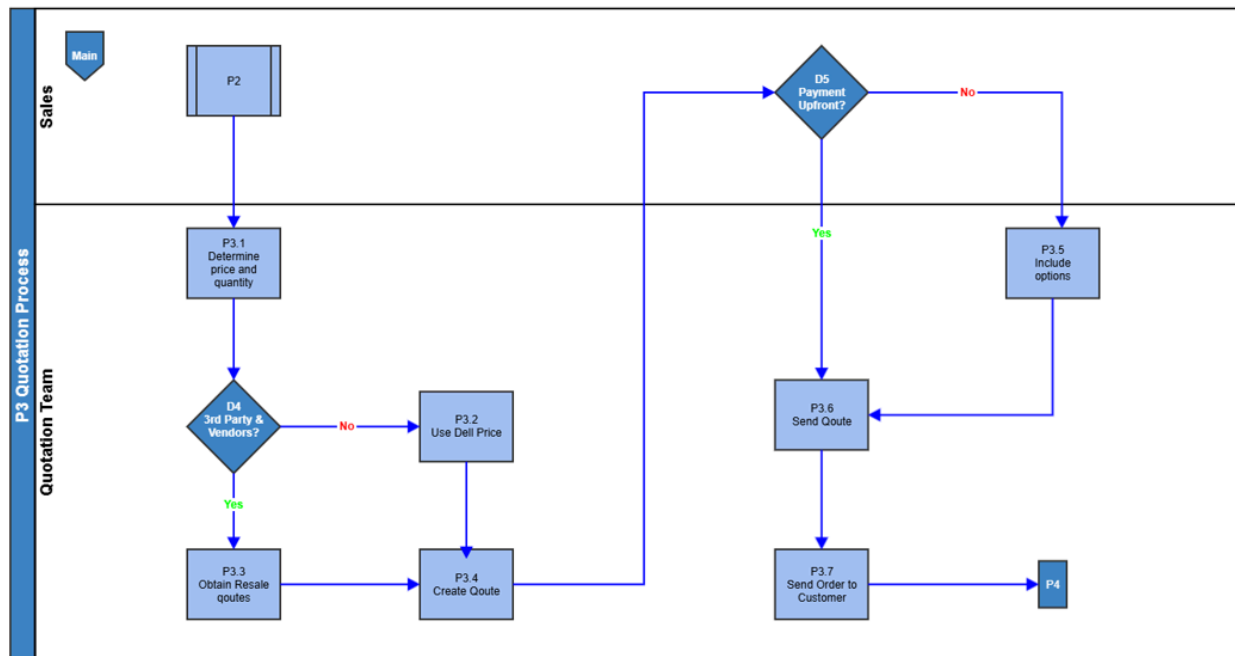
Baseline Business Process Model



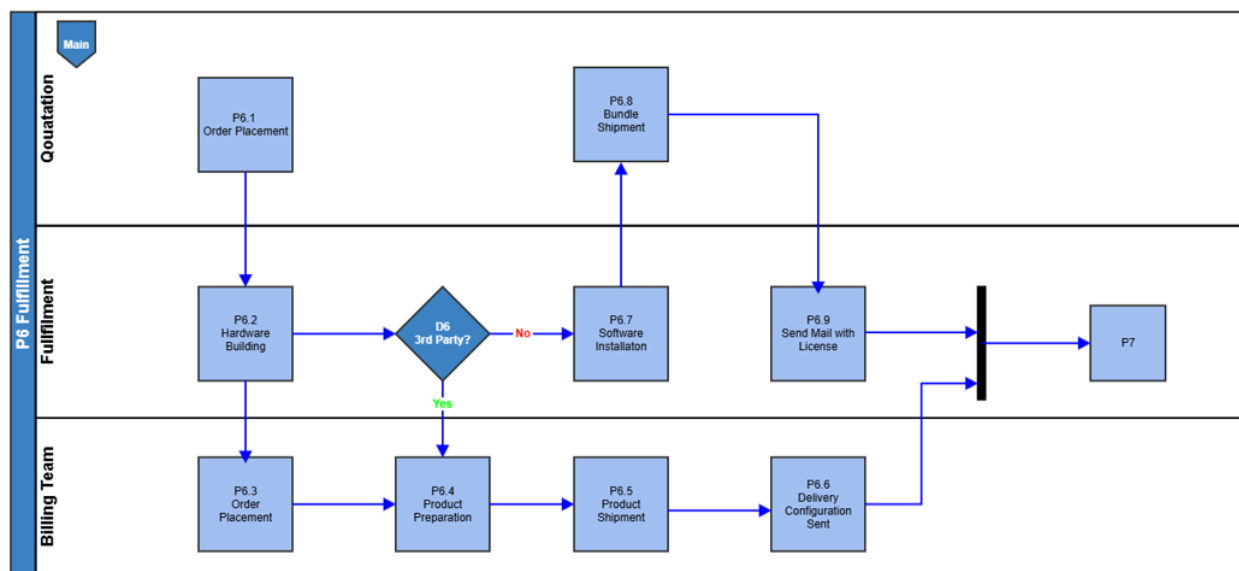
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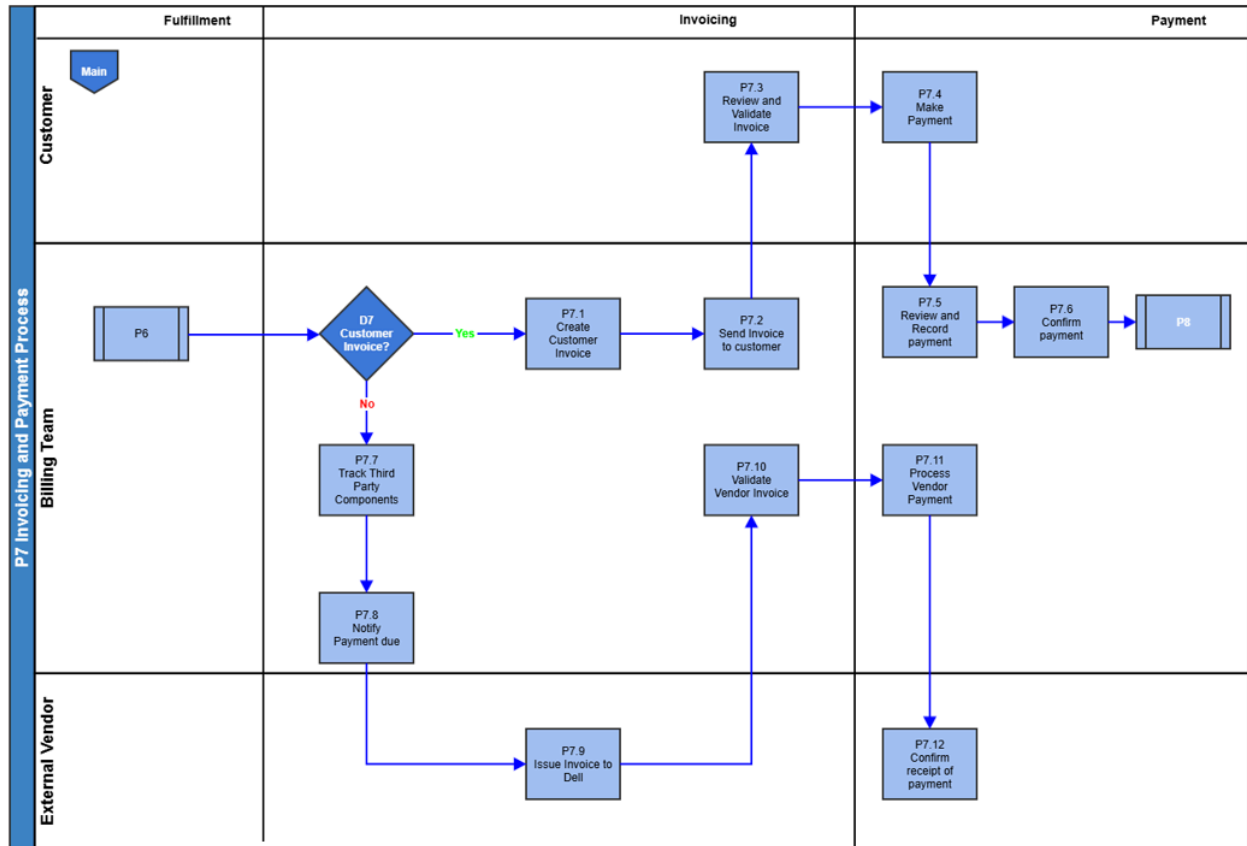
Quotation Process:



Fulfillment Process:

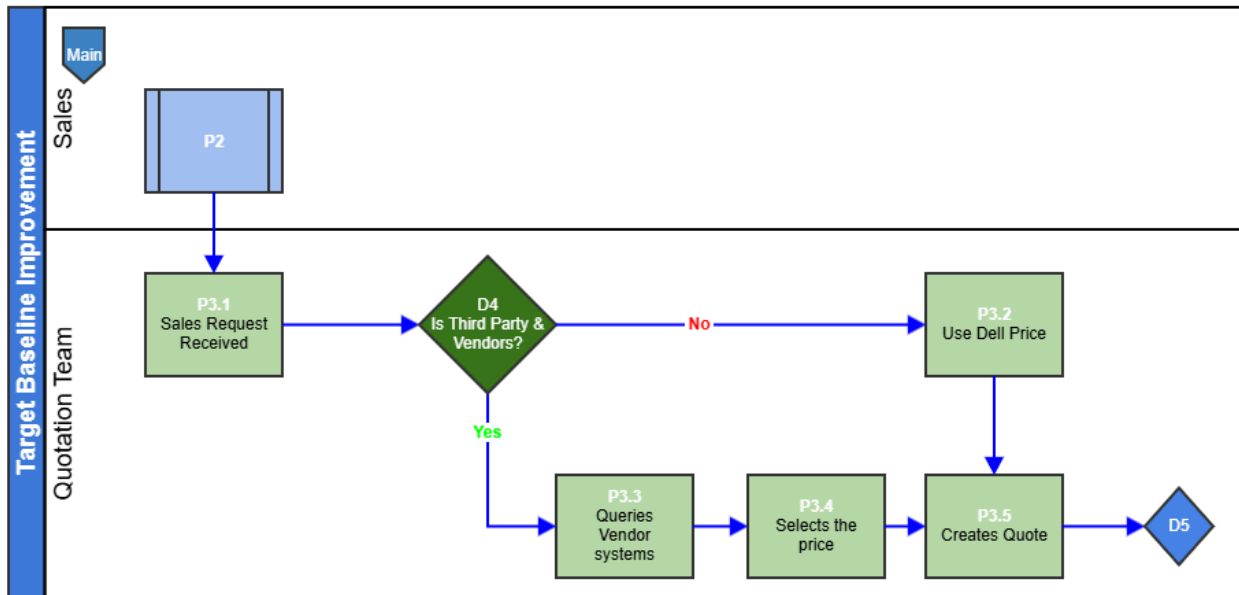


Invoicing and Payment Process:



Appendix B2

Target BPM Improvement



Appendix C

Actor Cards

<i>Actor Specification Card</i>	
Actor Name: Customer	
Type: Primary	Kind: Human
Role Description: Receives the Bill of Materials(BOM) and confirms or declines it. The customer will also interact with the system to submit the Purchase orders and make the payments for the product.	
Actor Goals: ● Receiving and Confirming the Bill of Materials ● Submitting the purchase order ● Making a payment	

<i>Actor Specification Card</i>	
Actor Name: Sales Executive	
Type: Primary	Kind: Human
Role Description: Creates Bill of Materials and sends them to the customer	
Actor Goals: ● Creating a Bill of Materials ● Sending The Bill of Materials to the customer	

<i>Actor Specification Card</i>	
Actor Name: Quotation Executive	
Type: Primary	Kind: Human
Role Description: Creates and sends quotations and sends purchase orders to the customer.	
Actor Goals: • Creating a Quote • Sending customer Quote • Sending order to the customer	

<i>Actor Specification Card</i>	
Actor Name: Billing Executive	
Type: Primary	Kind: Human
Role Description: Places orders for customers and creates and sends customer invoices. Also tracks the third-party components, and notifies customers of payments due.	
Actor Goals: • Placing Orders • Creating customer invoices • Sending invoices to the customer • Tracking third-party components • Notifying customers of payments that are due.	

<i>Actor Specification Card</i>	
Actor Name: Fulfillment executive	
Type: Primary	Kind: Human
Role Description: Receives purchase order placements on the system	
Actor Goals: • Receive order placements	

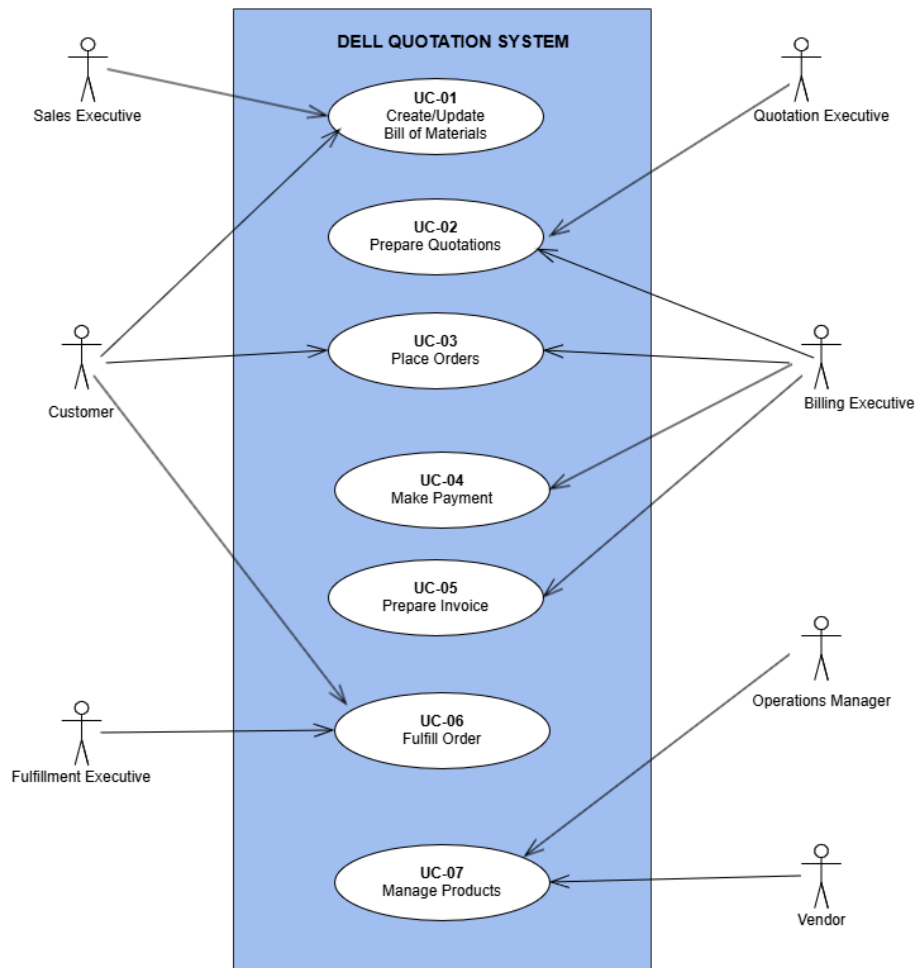
<i>Actor Specification Card</i>	
Actor Name: Vendor	
Type: Primary	Kind: Human
Role Description: Third-party software providers supplying products or services integrated into Dell's offerings or sold as add-ons.	
Actor Goals: • Provide accurate resale quotes for Dell's quoting process. • Deliver software licenses or services for customer fulfillment. • Receive timely payments for resold products/services.	

<i>Actor Specification Card</i>
Actor Name: Operations Manager

Type: Primary	Kind: Human
Role Description: Oversees and coordinates the integration and management of third-party vendor products and services within Dell's quote-to-cash process.	
Actor Goals: • Ensure smooth integration of third-party vendor systems and services. • Provide feedback on operational challenges and system improvements.	

Appendix D

Use Case Diagram



Appendix E

Use Cases: Functional Requirements

Use Case	UC-01. Create/Update Bill of Materials
Elaboration Phase	Initial Use Case
System Actors	Sales Executive, Customer
Business Process Steps/Decisions	P2
Business Process Description	After the sales executive has received the requirements from the customer, they log into the system to upload all the customer's hardware and software requirements for quotation purposes.
System Use Case Description	The Sales Executive logs into the system to upload the hardware requirements to the system. The sales executive then selects the specific vendors that they would like to match with the specific client.

Use Case	UC-02. Prepare Quotations
Elaboration Phase	Initial Use Case
System Actors	Quotation Executive, Customer
Business Process Steps/Decisions	P3, P5, D1
Business Process Description	The Quotation Executive uses the system to generate a quote based on the Bill of Materials provided by the Sales Executive, ensuring the quote includes accurate pricing and vendor details.
System Use Case Description	The Quotation Executive logs into the system, accesses the BOM that has already been populated by the system, and generates a quote that includes third-party vendor pricing. The executive reviews the quote and the system will automatically save it in the

	system. They will then send the order to the customer using the system.
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Use Case	UC-03. Place Orders
Elaboration Phase	Initial Use Case
System Actors	Customer, Quotation Executive, Billing Executive
Business Process Steps/Decisions	P4
Business Process Description	After generating the quote, the Quotation Executive sends it to the customer for review and potential approval.
System Use Case Description	The Quotation Executive selects the customer's information in the system, attaches the finalized quote, and sends it via the system's communication module to ensure secure and documented delivery.

Use Case	UC-04. Make Payment
Elaboration Phase	Initial Use Case
System Actors	Customer, Billing Executive
Business Process Steps/Decisions	P4
Business Process Description	The Customer reviews the quote, and if acceptable, submits a purchase order to initiate the order process.
System Use Case Description	The Customer logs into the system, views the quote, and submits a purchase order confirming the acceptance of terms and initiating order processing.

Use Case	UC-05. Prepare Invoice
Elaboration Phase	Initial Use Case
System Actors	Billing Executive
Business Process Steps/Decisions	P5
Business Process Description	The Billing Executive verifies the customer's purchase order against the generated quote and proceeds with order processing.
System Use Case Description	The Billing Executive accesses the submitted purchase order, verifies alignment with the quote, and confirms order processing within the system, marking it ready for invoicing.

Use Case	UC-06. Fulfill Order
Elaboration Phase	Initial Use Case
System Actors	Billing Executive, Fulfillment Executive
Business Process Steps/Decisions	P6
Business Process Description	Once the order is processed, the Billing Executive generates an invoice, sends it to the customer, and confirms payment to finalize the transaction.
System Use Case Description	The Billing Executive generates an invoice in the system, emails it to the customer using the system email module. The system will then send a payment link to the customer. After the customer has paid, the system will automatically update. The system logs all transaction details to maintain accurate financial records.

Use Case	UC-07. Manage Products
Elaboration Phase	Initial Use Case
System Actors	Vendor, Operations Manager
Business Process Steps/Decisions	P6
Business Process Description	Once the order is processed, the Billing Executive generates an invoice, sends it to the customer, and confirms payment to finalize the transaction.
System Use Case Description	The Billing Executive generates an invoice in the system, emails it to the customer using the system email module. The system will then send a payment link to the customer. After the customer has paid, the system will automatically update. The system logs all transaction details to maintain accurate financial records.

Appendix F

Data Model

